SALESTORM – STAGE 1

REQUIREMENT ANALYSIS & ASSUMPTIONS

High-Scale E-Commerce Flash Sale | SYSCRAFTERS 2026

1) PROBLEM STATEMENTS:

SALESTORM is a high-scale e-commerce flash-sale platform designed to handle sudden traffic spikes when a large number of customers attempt to purchase limited-stock products simultaneously. The system must support 10,000 concurrent purchase requests for 100 available units while preventing inventory overselling, duplicate reservations and duplicate payments. It must also maintain a reliable order lifecycle even when payment, database or downstream services experience delay or failures.

The system must:

- Handle sudden traffic spikes.
- Prevent inventory overselling.
- Reserve inventory temporarily.
- Process payments safely.
- Prevent duplicate transactions.
- Maintain a correct order lifecycle.
- Recover from service, database and payment failures.

Functional Requirements

1. **Product Discovery**
   - Users can view products and availability.
2. **Cart & Checkout**
   - Users can add products to cart and initiate checkout.
3. **Inventory Reservation**
   - Reserve available stock before payment.
   - Prevent two users from reserving the same unit.

4. **Payment**
    - Process successful and failed payments.
    - Prevent duplicate payments using idempotency.
5. **Order Management**
    - Create orders after successful payment.
    - Maintain valid order states.
6. **Reservation Expiry**
    - Release inventory when payment fails or reservation expires.
7. **Shipment & Notification**
    - Send confirmed orders for fulfilment and notify customers.

Non-Functional Requirements

Requirement	Target
Concurrency	10,000 simultaneous purchase requests
Inventory	Maximum 100 successful sales
Scalability	Horizontal scaling during peak traffic
Reliability	Recover from service/payment failures
Consistency	Inventory must never become negative
Performance	Handle sudden traffic spikes
Security	Authentication, HTTPS, rate limiting
Observability	Logs, metrics and distributed tracing